

MD ENAMUL HOQ

Ph.D. Candidate, Biomedical Informatics (Imaging) · Translational Imaging AI for Oncology

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[GitHub](#) · [LinkedIn](#) · [Google Scholar](#)

RESEARCH INTERESTS

Translational imaging informatics and computer vision for oncology, with emphasis on AI spanning radiology and digital pathology, large-scale imaging data pipelines, foundation models, and multimodal learning across CT and pathology. My work focuses on representation learning, DICOM-aware preprocessing, dataset curation and annotation, CT quality control, explainable AI, and clinically aligned evaluation — linking imaging phenotypes to histomorphological and molecular characterization to develop robust, interpretable, and reproducible systems for cancer diagnosis, monitoring, prognosis, and clinical translation.

EDUCATION

Ph.D., Biomedical Informatics (Imaging Track)

Expected May 2026

University of Arkansas for Medical Sciences (UAMS), Little Rock, AR

Dissertation: *Foundation Models for Lung Cancer Screening CT (LCS-DINO / Native-Resolution 2D).*

M.S., Engineering Physics

Dec 2020

Southeastern Louisiana University, Hammond, LA

Thesis: *Image Correlation for LDPE Film Deformation (computational imaging, MATLAB).*

B.S., Mechatronics Engineering (Biomedical Track)

Apr 2014

World University of Bangladesh, Dhaka

RESEARCH EXPERIENCE & APPOINTMENTS

Graduate Research Assistant (Ph.D. Researcher)

07/2021 – Present

University of Arkansas for Medical Sciences (UAMS), Little Rock, AR

- Built an end-to-end lung cancer screening (LCS) pipeline (data curation → embeddings → training → patient-level evaluation) for 20M+ CT slices, with full DICOM/PACS compliance.
- Managed large-scale cohorts in TCIA and the NCI Imaging Data Commons (IDC): cohort definition, manifest generation, DICOMweb/gsutil transfers, BigQuery/Python access, metadata harmonization (DICOM tags; RADLEX/SNOMED), and research governance (DUAs, PHI checks).
- Designed threshold-aware evaluation (ROC/AUC, PR/F1, optimal cut-points) to reflect real-world screening prevalence; reduced false-positive follow-ups in experiments.
- Scaled multi-GPU training and inference (A100) with Slurm; automated experiment tracking and report generation.
- Collaborated with 15+ clinicians and radiologists; translated findings into clinically interpretable metrics.
- Initiated and lead a longitudinal phantom-based study evaluating the reliability and drift of AI-based lung-nodule detection software using synthetic CT datasets at the Winthrop P. Rockefeller Cancer Institute, UAMS.
- Developed a pipeline to generate synthetic pulmonary nodules in DICOM CT volumes using MATLAB-based phantom modeling and PACS-derived retrospective imaging data.
- Designed a database and analysis workflow to evaluate detection accuracy, measurement reproducibility, and longitudinal model stability of commercial AI software (Riverain VS-CT).
- Collaborate with Radiology, Biomedical Informatics, and Cancer Institute investigators on an IRB-approved protocol for prospective validation of clinical AI systems.

Site Lead — Federated LDCT AI Collaboration

06/2025 – Present

UAMS, Little Rock, AR · with Moffitt, Vanderbilt, UCLA & U. Pittsburgh

- Lead Site 2 in a multi-institutional federated learning (FL) project for LDCT lung cancer screening using NVIDIA FLARE.
- Curate and harmonize multi-site imaging datasets; implement leakage-safe splits and AI-ready pipelines across institutions.

- Design and deploy privacy-preserving training workflows (secure model updates, cross-site evaluation protocols) without sharing raw PHI.
- Work with radiologists and data-science teams to improve detection performance and reduce clinical workload in real-world screening settings.

AI & Informatics Research Intern

02/2025 – 07/2025

Mayo Clinic, Rochester, MN

- Developed a multimodal pathology retrieval and auto-captioning framework using vision foundation models and LLMs — Top-Ten Poster, Mayo Clinic AI Summit 2025.
- Improved semantic alignment of generated captions (>0.82 cosine similarity) using hybrid rule-based + LLM summarization pipelines.
- Co-led a workshop on fine-tuning and deploying Vision-LLMs in clinical research workflows.

Graduate Research & Teaching Associate; Lab Manager

08/2019 – 12/2020

Southeastern Louisiana University, Hammond, LA

- Developed MATLAB-based pipelines for digital image correlation (LDPE deformation) with visualization and modeling tools.
- Supported 13+ researchers with debugging and optimization; instructed Physics 101/102 laboratory courses.

Research Intern — Multimedia Lab

09/2018 – 02/2019

National Chung Cheng University, Chiayi, Taiwan

- Developed depth-estimation algorithms for 2D → 3D video conversion, improving perceived depth and visual quality.

R&D Engineer

08/2017 – 07/2018

TN Solution — Karlsruhe, Germany & Ho Chi Minh City, Vietnam

- Led image/signal-processing and IoT projects; improved PCB fault diagnosis using noise reduction and feature-extraction techniques.

PUBLICATIONS & SCHOLARLY WORK

Peer-Reviewed Publications

1. **Hoq ME**, Tarbox L, Johann D Jr, Larson-Prior L, Prior F. Harnessing Native-Resolution 2D Embeddings for Lung Cancer Classification: A Feasibility Study with the RAD-DINO Self-Supervised Foundation Model. *Journal of Imaging Informatics in Medicine*. 2025 (pp. 1–23).
2. Alfasly S, Uegami W, **Hoq ME**, Alabtah G, Tizhoosh HR. Semantic & Visual Crop-Guided Diffusion for Heterogeneous Tissue Synthesis in Histopathology. *Advances in Neural Information Processing Systems (NeurIPS)*. 2025.

Accepted / In Press

1. **Hoq ME**, Larson-Prior L, Prior F. Virtual-Eyes: Quantitative Validation of a Lung CT Quality-Control Pipeline for Foundation-Model Cancer-Risk Prediction. *Medical Imaging with Deep Learning (MIDL)*. Accepted, 2026.
2. **Hoq ME**, Uegami W, Alfasly S, Alabtah G, Malakshan S, Kazemi A, Schmitgen AT, Prior F, Tizhoosh HR. Retrieval-Guided Generation for Safer Histopathology Image Captioning. *Ital-IA*. Accepted, 2026.

Preprints & Technical Reports

1. **Hoq ME**, Prior F. Artificial Intelligence for Low-Dose CT Lung Cancer Screening: A Systematic Review of Deep Learning, Input Dimensionality, Workflow Integration, and Foundation Models. *Preprints.org*, 2026. Posted 16 June 2026 (v1; not peer-reviewed). doi:10.20944/preprints202606.1208.v1.
2. Mu J, Karna AK, Kumar Koushik T, Winchell J, Yang TJ, Zhang CM, Baker J, Hagen E, **Hoq ME**, Kim K, ... Busby B. Towards Federated Learning Across Biobanks: Prototype Software from the 2026 Carnegie Mellon University–NVIDIA Hackathon. *OSF Preprint*, 2026.

Published Abstracts & Conference Proceedings

1. **Hoq ME**, Uegami W, Alfasly S, Malakshan SR, Alabtah G, Kazemi A, et al. From CNN to Vision Foundation Models and LLMs: A Multimodal Framework for Pathology Image Retrieval and Auto-Captioning. *Mayo Clinic Proceedings: Digital Health*, 3(4), 2025 (Abstract).
2. Kolb S, Sunkara H, Chavane C, **Hoq ME**, Ali M, Xu H, Kashyap R. Renal Replacement Therapy Following IV Contrast Administration in Critically Ill Patients with CKD. *Critical Care Medicine*, 53(1), 2025 (Abstract).

3. Barnes B, **Hoq ME**, Binette E, Lee S, Arumairaj A. Impact of an Elevated Fluid-Accumulation Index in Septic Patients with HFREF: A MIMIC-IV Analysis. *Critical Care Medicine*, 2026 (Abstract).
4. **Hoq ME**, Prior F. Optimizing and Harnessing 20 Million NLST CT Lung-Screening Images for Robust Foundation-Model Training. *NIH Early Detection Research Network (EDRN)*, 2024 (Abstract).
5. **Hoq ME**, Yoshida S, Jung H. Measurement of Thickness and Preservation of Thin Films at Maximum Wrapping Speed. *APTEC Annual Meeting*, 2019.

Thesis

1. **Hoq ME**. An Investigation of Image Correlation for Real-Time Monitoring of Wrapping-Film Deformation. *Master's Thesis*, Southeastern Louisiana University, 2020.

SELECTED INVITED TALKS & POSTERS

- **UAMS Radiology Retreat, 2026** — Synthetic CT Phantom for Nodule-Detection Validation.
- **UAMS Winthrop P. Rockefeller Cancer Institute, 2026** — From CT Quality Control to Structured Radiology Reporting & Validation.
- **NCI Early Detection Research Network (Caltech), 2024** — Optimizing & Harnessing 20M NLST CT Lung-Screening Images.
- **APTEC Annual Meeting, 2019** — Digital Image Correlation for LDPE Deformation (audience 200+).

GRANTS & LARGE-SCALE RESEARCH INITIATIVES (CONTRIBUTOR)

- **NCI U24 CA215109 — The Cancer Imaging Archive (TCIA) Sustainment & Scalability**. Contributor to large-scale imaging-data infrastructure supporting multi-institutional AI research and data sharing.
- **UAMS Team Science Award (\$100K)**. Lead and main contributor to an interdisciplinary initiative focused on clinical AI and imaging analytics.
- **UAMS Team Science Award (\$40K)**. Lead and main contributor to a collaborative program supporting translational imaging and data-driven clinical studies.
- **EU EuCanImage (SEP 210631720; €805,479)**. Contributor to an international imaging-AI ecosystem for multi-center data integration and cancer research.
- **MOTIE Korea International R&D Program (P0006842)**. Contributor to global technology collaboration in imaging and data-driven innovation.

HONORS & AWARDS

- Champion (3 of 5 Categories) — CMU–NVIDIA Federated Learning Hackathon, 2026
- Top-Ten Poster — Mayo Clinic AI Summit, 2025
- Second Place — Datathon.org (Emory University School of Medicine), 2024
- Travel Grants — SIIM (2025), SCCM (2024–2025), HITLAB (2024)
- Merit Scholarship — World University of Bangladesh

PROFESSIONAL SERVICE & LEADERSHIP

Editorial & Peer Review

- Ethics Reviewer — NeurIPS 2026
- Program Reviewer — Society for Imaging Informatics in Medicine (SIIM) Annual Conference, 2025–present
- Ad Hoc Reviewer — Frontiers in Medicine, 2024–present (6+ manuscripts reviewed)
- Ad Hoc Reviewer — Journal of Thoracic Disease
- Poster Judge — UAMS Summer Research Program, Winthrop P. Rockefeller Cancer Institute, 2024 (17 posters evaluated)

Conference & Project Leadership

- Area Chair (Challenges Track) — Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026
- Site Lead (UAMS) — Multi-institutional Federated Learning Project for LDCT Lung Cancer Screening (Moffitt, Vanderbilt, UCLA, U. Pittsburgh)
- Workshop Co-lead — Practical AI in Healthcare: Fine-tuning & Deployment with Open-Source Tools, Mayo Clinic, 2025

TEACHING & MENTORING

- **Research Mentorship — Medical Imaging AI, UAMS (2023–present).** Mentored 3 Ph.D. students and 1 M.D. student in medical-imaging AI, including GPU/HPC workflows, DICOM/PACS data handling, experimental design, and reproducible research practices.
- **Graduate Teaching & Research Associate — Southeastern Louisiana University (2019–2020).** Instructed Physics 101/102 laboratories and lectures; supported curriculum delivery and student training in experimental methods.
- **Instructor & Mentor — Foundation-Year Physics & Mathematics (2010–2017).** Taught and mentored 10,000+ pre-university students across multiple cohorts; developed problem-based learning materials and structured exam-preparation frameworks.

TECHNICAL SKILLS

Programming: Python, C/C++, Java, MATLAB

Frameworks & Libraries: PyTorch, TensorFlow, Keras

Medical Imaging & Standards: DICOM, PACS, RADLEX, SNOMED, TCIA, TCGA, 3D Slicer, ITK-SNAP

Compute & Infrastructure: Linux, Slurm, HPC (A100 multi-GPU), Google Cloud Platform (GCP), AWS

Data Engineering & Evaluation: Leakage-safe dataset splitting, imbalanced learning, ROC/AUC/PR/F1 analysis, calibration and threshold optimization, BigQuery (IDC), TCIA manifests, DICOMweb pipelines

CERTIFICATIONS

- Certified in Lung Cancer Screening Education — American College of Radiology (ACR), 2024
- National Imaging Informatics Course (NIIC) — SIIM / AAPM / RSNA
- Professional Certifications (20+) — NVIDIA, Google, Coursera (AI, Deep Learning, Cloud Computing, Data Science)

SELECTED DATASETS & SOFTWARE

- **NLST-Derived CT Screening Cohort (20M+ slices).** Internal UAMS-curated dataset: preprocessing, DICOM normalization, leakage-safe patient-level splits, and RAD-DINO embedding pipelines for large-scale model training and evaluation.
- **Multimodal Pathology Retrieval & Auto-Captioning Pipeline.** Vision foundation-model embeddings + rule-based labeling + LLM summarization; interactive analysis tools for retrieval evaluation and semantic alignment.
- **HunCRC & ARCH Datasets.** Generated embeddings and performed leave-one-out retrieval analysis for histopathology image similarity and representation benchmarking.

PROFESSIONAL MEMBERSHIPS

- The MICCAI Society (Medical Image Computing and Computer Assisted Intervention)
- Society for Imaging Informatics in Medicine (SIIM)
- Society of Critical Care Medicine (SCCM)
- Radiological Society of North America (RSNA)
- American College of Radiology (ACR)

REFERENCES

- **Prof. Dr. Fred W. Prior** — Distinguished Professor & Chair, DBMI, UAMS · FWPrior@uams.edu · +1 (314) 303-2485
- **Dr. Lawrence Tarbox** — Associate Professor & Head of HPC, DBMI, UAMS · lrtarbox@uams.edu · +1 (501) 603-1766
- **Dr. Donald J. Johann Jr.** — Professor (M.D.), Cancer Therapeutics, UAMS · djjohann@uams.edu · +1 (501) 526-6990